# NEF ABUNDANCE: FRAMEWORK

## 1. Theorem and proof

**Theorem.** *Let $X$ be a smooth projective threefold over $\mathbb{C}$ such that $K_X \sim 0$ and $q(X) = 0$. Let $D$ be a nef Cartier divisor on $X$ such that $\mathrm{n}(X, D) = 3$ and $D^3 = 0$. Then*

$$H^0(X, \mathcal{O}_X(mD)) \neq 0 \qquad \text{for some integer } m > 0.$$

*Proof.*

**Setup.**

- Fix $\epsilon \in \mathbb{Q}_{>0}$ sufficiently small such that

$$343\epsilon^3 < \tfrac{1}{4}, \qquad 40\epsilon < \tfrac{3}{8}.$$

- Fix an ample Cartier divisor $A$. For integers $n \geq 1$, put

$$a_n = ((D + A/n)^3)^{1/3}, \qquad r_n = \max\{1, \lfloor \epsilon/a_n + 1/2 \rfloor\}, \qquad P_n = r_n(D + A/n).$$

  Then $P_n$ is ample and rational Cartier, $r_n \to \infty$, and $P_n^3 \to \epsilon^3$. Indeed, $a_n \to 0$ and $|r_n a_n - \epsilon| \leq a_n/2$ for all sufficiently large $n$.

- Let $p_i : X \times X \to X$ be the projections, for $i = 1, 2$. Put

$$Z = \mathbb{P}_{X\times X}\big(p_1^*\mathcal{O}_X(D) \oplus p_2^*\mathcal{O}_X(D)\big).$$

  Let $\pi : Z \to X \times X$ be the projection. Let $S_1, S_2 \subset Z$ be the two summand sections and set

$$U = Z \setminus (S_1 \cup S_2).$$

- Set $\xi = c_1(\mathcal{O}_Z(1))$. For a positive integer $s$, put

$$R_{n,s} = (p_1 \circ \pi)^* P_n + (p_2 \circ \pi)^* P_n + s\xi.$$

  Whenever $|kR_{n,s}|$ is used, $k > 0$ is an integer with $kR_{n,s}$ Cartier, equivalently $kP_n$ Cartier.

First, we show the following result.

**Lemma 1.1.** *There exists a positive integer $s$ such that, for every $n \gg 0$, $|kR_{n,s}|$ separates $8k$-jets at some $z \in U$ for some integer $k > 0$ with $kR_{n,s}$ Cartier.*

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Suppose for contradiction that $\kappa(D) = -\infty$. We show that the global sections of $kR_{n,s}$ do not generate all $8k$-jets, as shown in the next lemma.

**Lemma 1.2.** *For every positive integer $s$ there exists an integer $n_0 = n_0(\epsilon, s) \geq 1$ such that, for every integer $n \geq n_0$, $|kR_{n,s}|$ does not separate $8k$-jets at any $z \in U$, for any positive integer $k$ with $kR_{n,s}$ Cartier.*

Together, Lemmas 1.1 and 1.2 give a contradiction for the same $s$ and sufficiently large $n$. □

**Draft status.** The proofs of the two geometric lemmas are not included or independently verified here. Lemma 1.2 combines the earlier small-functional input and the jet-obstruction input; it is not a newly verified result. This draft is for discussion and checking.